Case Study: Multi-Dimensional Modeling of Local Retail Performance

Strategic Market Positioning through Customer Sentiment & Engagement Analytics

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Abstract

This research addresses the "Small Sample Bias" prevalent in digital retail reputation systems within the Bangalore ecosystem. By analyzing a dataset of 59 unique retail units, this study introduces a Weighted Popularity Score utilizing logarithmic normalization. The model successfully rewards quality at scale, providing a reproducible framework for distinguishing true market anchors from statistically insignificant outliers. The technical lifecycle—spanning Excel-based engineering, R-based statistical validation, and Tableau-driven visualization—is documented herein.

Nomenclature

R	Raw Star Rating (0.0 – 5.0)
v	Customer Review Volume (Total Engagements)
P	Weighted Popularity Score (The Engineered Index)
$\log_{10}$	Common Logarithm (Base-10 Normalization Factor)

1. Introduction

In the contemporary retail landscape of Bangalore, digital reputation acts as the primary catalyst for customer acquisition. However, raw star ratings are frequently misleading due to statistical volatility in small datasets. This study aims to move beyond surface-level metrics to engineer a system that normalizes quality across varying scales of engagement.

2. Problem Statement: The Small Sample Bias

The fundamental credibility gap identified in this research is the lack of volume-weighting in standard ranking systems. A business with a single 5-star rating mathematically appears superior to a market leader possessing a 4.5-star rating across 1,000 customers. This project addresses this distortion by engineering a metric that acknowledges the non-linear relationship between consumer trust and volume.

3. System Architecture and Technical Stack

The implementation followed a modular three-stage technical pipeline to ensure data integrity:

- **Data Engineering (Microsoft Excel):** Standardization of 11 unique service types and 6 core business categories (Boutique, Pharmacy, Salon, Supermarket, Kirana, and F&B).
- **Statistical Validation (R-Statistical Computing):** Application of the R-Markdown framework (`Shop_Analysis.Rmd`) for correlation testing and model verification.
- **Business Intelligence (Tableau):** Development of an interactive dashboard (`shops_dashboard.twb`) for performance quadrant mapping.

4. Mathematical Modeling

The core intellectual property of this study is the **Weighted Popularity Score** (P), defined as:

$$P = R \cdot \log_{10}(v + 1) \tag{1}$$

The base-10 logarithm recognizes that the jump from 1 to 10 reviews represents a greater leap in initial credibility than moving from 1,000 to 1,010. This damping ensures that market anchors are rewarded for their established "Trust Equity" without making niche shops invisible.

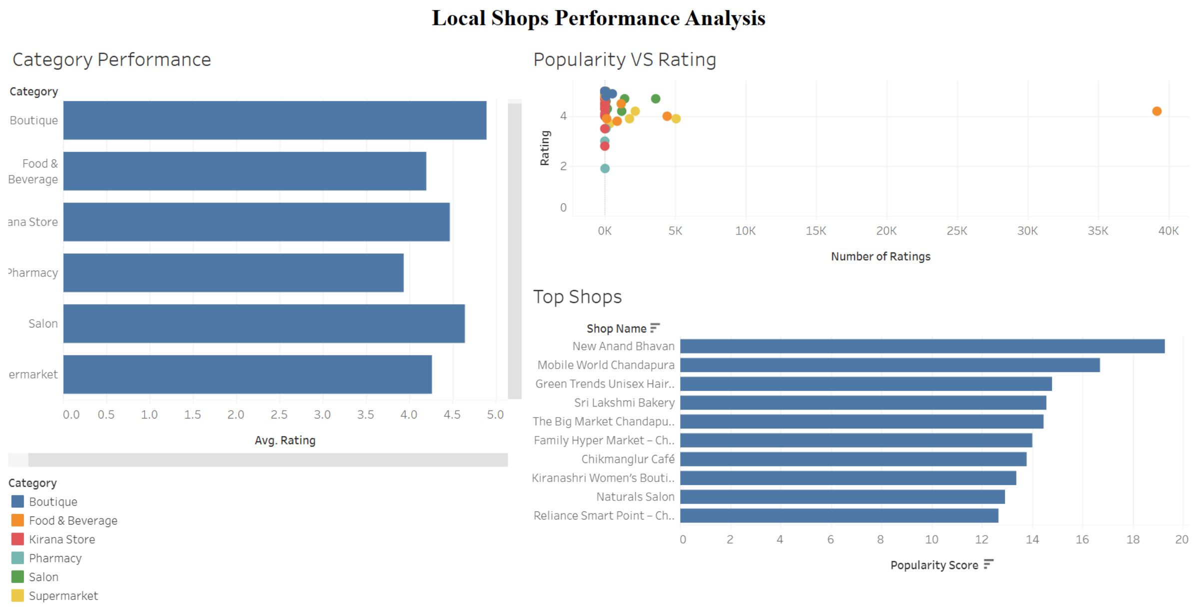

Figure 1: Integrated System Dashboard: Multi-Dimensional Retail Performance Overview

5. Analytical Results and Sector Benchmarking

5.1 Quality Consistency by Sector

Analysis revealed significant quality variance between sectors. The Boutique sector serves as the benchmark leader in Bangalore for quality consistency (Avg: 4.9/5.0), while the Pharmacy sector exhibited the lowest average ratings (3.9/5.0).

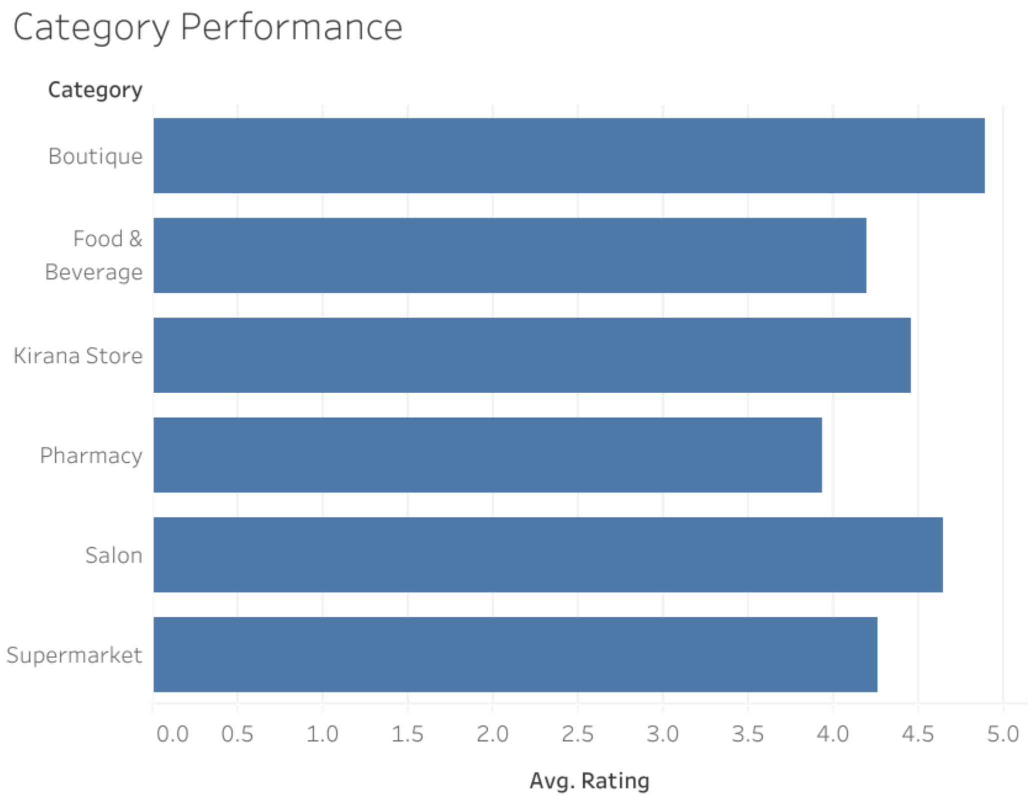

Figure 2: Sector Benchmarking: Average Rating Analysis across 59 Units

5.2 Strategic Market Positioning Quadrants

By mapping businesses on a Rating (R) vs. Engagement (v) scale, four strategic tiers were identified:

1. **Tier 1: Established Leaders:** High Rating + High Volume (e.g., New Anand Bhavan).
2. **Tier 2: Hidden Gems:** High Rating + Low Volume; elite service lacking digital discovery.
3. **Tier 3: Volume Challengers:** High Volume + Lower Quality.
4. **Tier 4: At-Risk Units:** Low Quality + Low Volume.

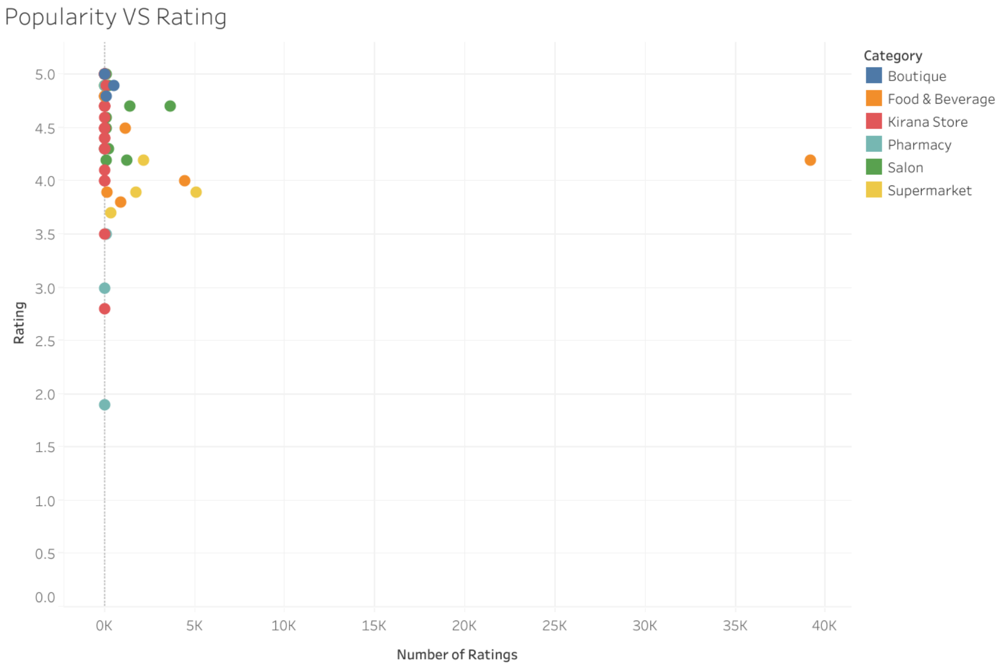

Figure 3: Strategic Tier Mapping: Quality consistency against Consumer Reach

5.3 Identification of Market Authority

The Popularity Leaderboard provides a ranking that prioritizes brand reliability over raw footfall. Leaders such as *Green Trends* and *Mobile World Chandapura* rank high because they successfully grew their customer base without letting service standards slip.

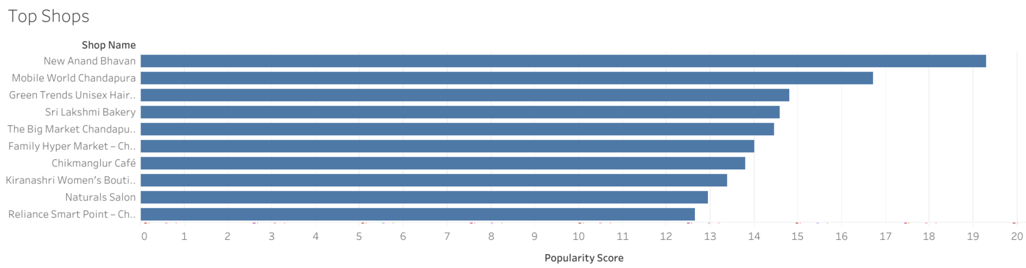

Figure 4: Market Health Leaderboard: Top Performing Units by Popularity Score

6. Conclusion

This research concludes that in a competitive ecosystem, Quality is the floor, but Engagement is the ceiling. By applying logarithmic normalization, we provide Bangalore's local business owners with a realistic KPI to track their market authority relative to high-volume competitors.

Project Artifacts for Verification

- `Shop_Cleaned.xlsx` (Cleaned Dataset)
- `Shop_Analysis.tex` (Technical Documentation)
- `shops_dashboard.twb` (Interactive BI Suite)